

Figure 4 — absolute calibration products (median $\pm$ robust σ)

stage	APEX	Python ccdproc	IRAF ccdproc
master bias	512 $\pm$ 1.483	512 $\pm$ 1.483	512 $\pm$ 1.483
master dark (60 s)	1 $\pm$ 2.224	1 $\pm$ 2.224	1 $\pm$ 2.224
master flat	1 $\pm$ 0.04428	1 $\pm$ 0.04428	1.001 $\pm$ 0.04431
science after full chain	633.5 $\pm$ 31.39	633.5 $\pm$ 31.39	633 $\pm$ 31.37

All entries are absolute output values; no APEX-reference subtraction is plotted.

Master flat is unit-median normalized. Robust $\sigma = 1.4826 \times \text{MAD}$ over the full frame.

NGC 6811 B 60 s · Moravian C3-61000 (2×2) · 8 bias / 8 dark / 5 flat · NGC 6811 B, 60 s, 2026-06-11